E49 review pack: Fig. 4g in two versions

For the author at STOP 3. Exploratory and post hoc.

2026-11-03

What this is. One section per version of Fig. 4g, each starting on a new page:

- **the panel at final size**, 178 mm wide;
- **the text**: brief section 8, finalised against the verified numbers, with the amendment's two sentences and the effector sentence;
- **the legend draft**, verbatim from the script, with the § note, followed by the proposed additions (brief 5.6), marked as such;
- **STYLE.md metrics** for every paragraph.

The Extended Data panels follow the two versions.

What is the same in both versions.

- **The values.** Every cell draws the same coefficient; the script asserts it. Version (b) adds a dot on each gene cell whose 95% CI excludes zero.
- **The text and the metrics.** Only the legend gains one line in version (b).
- **The choice carries** to ED-e, ED-f and ED-h, which are shown here in both versions.

The record.

- **Source:** `docs/2026-11-03_e49_result.md`. Part 4 is this text, and its section 4.1 says what changed from the brief's drafts and why.
- **The run:** 24 of 24 digests and 192 of 192 assertions; 33 of 33 PNGs and tables byte-identical to the dry run. Built 2026-11-03 13:05:54 GMT.

One qualification found after the run (result note, Part 2 H2; AD HOC arithmetic, not declared).

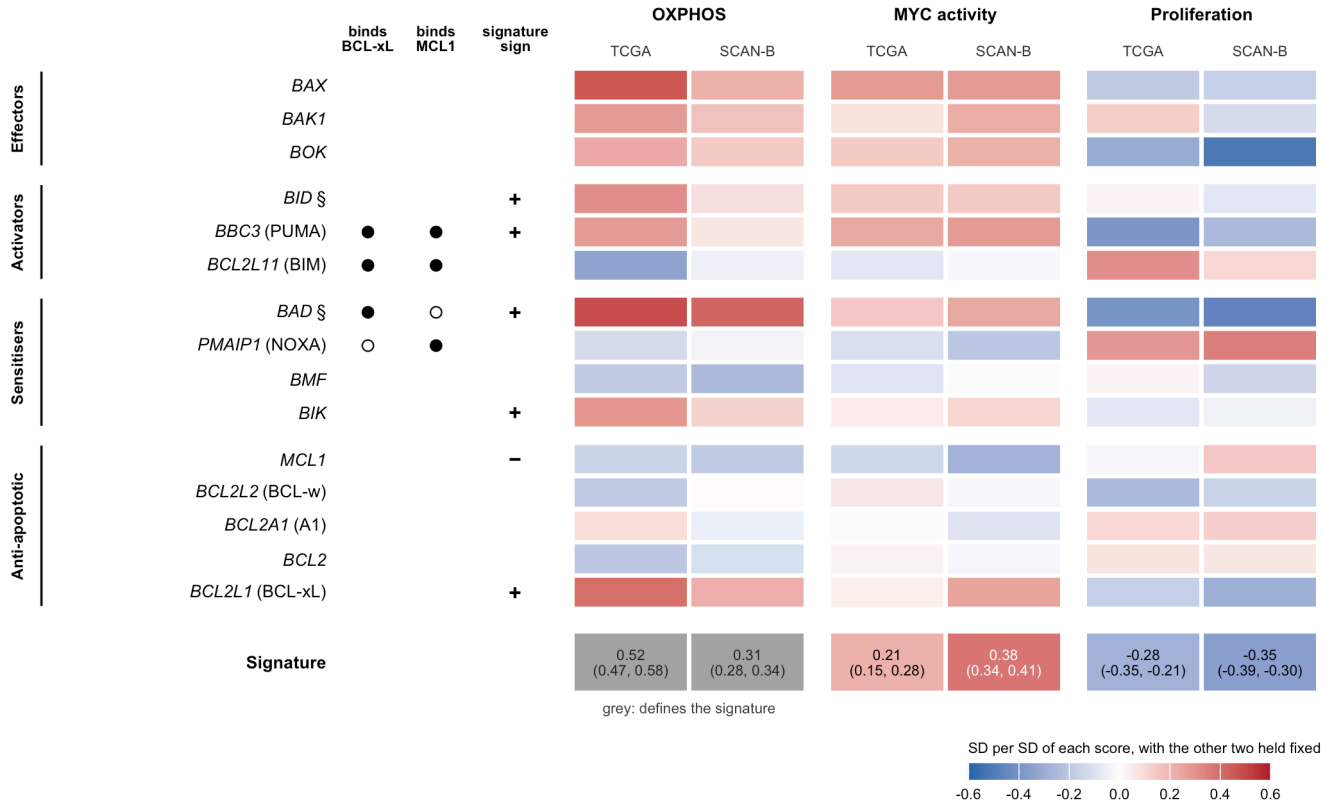
- **The issue.** Part of the signature's proliferation coefficient can be inherited from its selection rule, which chose genes by their correlation with OXPHOS at fixed proliferation.
- **How much.** If all of the signature's OXPHOS coefficient came from that selection, the inherited part would be up to about half of the coefficient on 4g's model in TCGA, and a third in SCAN-B.
- **What remains.** The remainder is negative in every cell, but it has no interval.
- **MYC activity's coefficient is unaffected.**
- **Where it bears on the text:** S5 ("played no part in its definition"), S6, S8, A2 and Z1. The test would be E50.

How the metrics are counted. Words are whitespace tokens containing a letter or digit. Figure citations in parentheses and reference numbers are not counted.

Decisions for the author.

1. **The version:** (a) unmarked, or (b) marked.
2. **The choices marked "the author's call"** in the result note, 4.1:
   - S3 and S4b, the six-gene sentence;
   - S5;
   - the abstract (A1);
   - the closing sentence (Z1);
   - "rely on" (M3);
   - "consequences".
3. **Whether to declare E50**, the test of H2, before the proliferation sentences are final.
4. **The proposed legend additions.**
5. **The binding code for BID, BIK and BMF**, which is drawn blank until the author supplies it.

Version (a): unmarked



The text

Abstract: the human sentences. Extending this relationship to 4,081 human breast tumours[8,9], an OXPHOS-associated BCL-2 family signature showed BCL-xL rising more steeply than PUMA. MYC activity, once separated from proliferation, was associated with the same signature, whereas proliferation was associated with the opposite when OXPHOS was held fixed.

Mouse Fig. 4 paragraph: last sentence (unchanged). Consequently, established human tumours with high OXPHOS and MYC activity were predicted to carry raised pro-apoptotic signalling restrained by BCL-xL, distinct from the consequences of proliferation.

The signature paragraph. In tumours, however, OXPHOS rises with both MYC activity and proliferation, so their associations can only be interpreted once the three are separated. In TCGA-BRCA (n = 938) and SCAN-B (n = 3,143) we therefore modelled each BCL-2 family gene on all three together, adjusting for intrinsic subtype (Fig. 4g). Two of four MYC activity scores, including the widely used Hallmark set, correlated with proliferation too closely to be separated from it, so we used one containing few proliferation genes (Extended Data Fig. Xa). Six of the twelve BH3-only and anti-apoptotic genes, the mitochondrial members most closely associated with OXPHOS, formed a signature (Fig. 4g). BCL2L1, encoding BCL-xL, rose with the activators BBC3 and BID and the sensitisers BAD and BIK, whilst MCL1 fell. Outside the signature, BCL2L11, BMF and BCL2 also fell with OXPHOS in both cohorts (Fig. 4g). Because the signature was defined by OXPHOS, that association is descriptive, whereas its associations with proliferation and MYC activity played no part in its definition. Although proliferation correlated with OXPHOS (ρ = 0.23–0.42; Extended Data Fig. Xa), the signature did not follow it (Extended Data Fig. Xh), and at fixed OXPHOS and MYC activity higher proliferation went with lower signature expression (Fig. 4g). On the score with few proliferation genes, and on a second, equally separable score, MYC activity went with the signature in the same direction as OXPHOS, whereas the other two scores showed this less (Fig. 4g and Extended Data Fig. Xj). OXPHOS and MYC activity therefore went with the signature in one direction, and proliferation in the other. The effectors BAX, BAK1 and BOK, outside the signature, rose with OXPHOS and MYC activity in both cohorts, whilst proliferation went with lower BAX and BOK (Fig. 4g).

The balance paragraph. Within the signature, BCL2L1 rose more steeply with OXPHOS than BBC3, encoding PUMA, in both cohorts (Extended Data Fig. Xb). This is the reverse of the MYC-expressing gland, in which PUMA rose relative to BCL-xL with OXPHOS priority (Fig. 1i; Extended Data Fig. Xk), and resembles the mouse tumours that sustained high OXPHOS only alongside Bcl-xL (Fig. 3h,i). BAD rose most steeply of all (Fig. 4g), but BAD neutralises BCL-xL only when unphosphorylated, and survival signals phosphorylate it and hold it away from BCL-xL[Zha]. Its expression therefore does not measure its activity, unlike that of PUMA, which is controlled mainly through transcription[Nakano].

The MCL1 paragraph. MCL1 fell as BCL-xL rose, so BCL-xL was the only one of the two main guardians to rise alongside PUMA, which binds both[Chen]. MCL1′s reported support of respiration in breast cancer stem cells depends on a mitochondrial function separable from its anti-apoptotic role[27]. This suggests that tumours with high OXPHOS rely on BCL-xL rather than MCL1, like persistent OXPHOS-high triple-negative breast cancer cells, which depend on BCL-xL and are insensitive to MCL1 inhibition[26].

The clinical paragraph: edited sentences only. Having separated OXPHOS from proliferation on the BCL-2 family, we next asked whether the two are also separable in their association with chemotherapy response. Additional adjustment for MYC activity, and then for

the mean expression of BCL2L1 and MCL1, did not materially alter the OXPHOS–response association (pooled odds ratios 0.87 and 0.88). The transcriptional coupling of OXPHOS and proliferation therefore concealed a clinically relevant distinction.

The closing paragraph: its human sentence. In human breast tumours, OXPHOS and MYC activity were associated with a BCL-2 family signature in which BCL-xL outpaced PUMA, whereas proliferation ran the other way, on this signature as on chemotherapy response.

Methods notes (the signature's definition is E47 declaration 4a.7, item 5, unchanged and not re-measured here). The effectors BAX, BAK1 and BOK were fitted for comparison with the groups of Fig. 1g and were not added to the signature. Binding of BCL-xL and MCL1 by each BH3-only protein was taken from published peptide affinities and is shown only where the source states it[Chen]. This analysis was post hoc, and the signature's proliferation coefficient had been seen in an earlier analysis before it was declared; no reading rule was applied, and every coefficient is reported.

Legend draft

Legend draft, verbatim from the script (191 words). Associations of OXPHOS level, MYC activity and proliferation with the expression of fifteen BCL-2 family genes, in the groups of Fig. 1g, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each row is one linear model of the gene's expression on OXPHOS, MYC activity and proliferation, with intrinsic subtype, and with tumour purity and leukocyte fraction in TCGA-BRCA; each coefficient is in SD per SD with the other two held fixed, and every variable is rank-normalised within each cohort. MYC activity is scored on MYC_UP.V1_UP, a score with few proliferation genes; the other MYC activity scores are shown in Extended Data. Left, binding of BCL-xL and MCL1 by BH3-only proteins, from Chen et al. (2005): filled, binds; open, does not; blank, not yet verified. Signature sign: the direction in which each of the six signature genes enters the signature. § BH3-only proteins whose activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). Bottom, the six-gene signature: its coefficient and 95% confidence interval in each column; its association with OXPHOS, in grey, defines the signature and is shown, not interpreted. OXPHOS priority is shown in Extended Data.

Proposed additions, not in the script's draft (brief 5.6; result note 4.3) (42 words). Proliferation went with higher BAK1 in TCGA-BRCA and lower BAK1 in SCAN-B, and with higher MCL1 in SCAN-B only. BCL2L11 (BIM) fell with OXPHOS more steeply in TCGA-BRCA than in SCAN-B. PMAIP1 (NOXA) and BCL2L11 went with higher proliferation in both cohorts.

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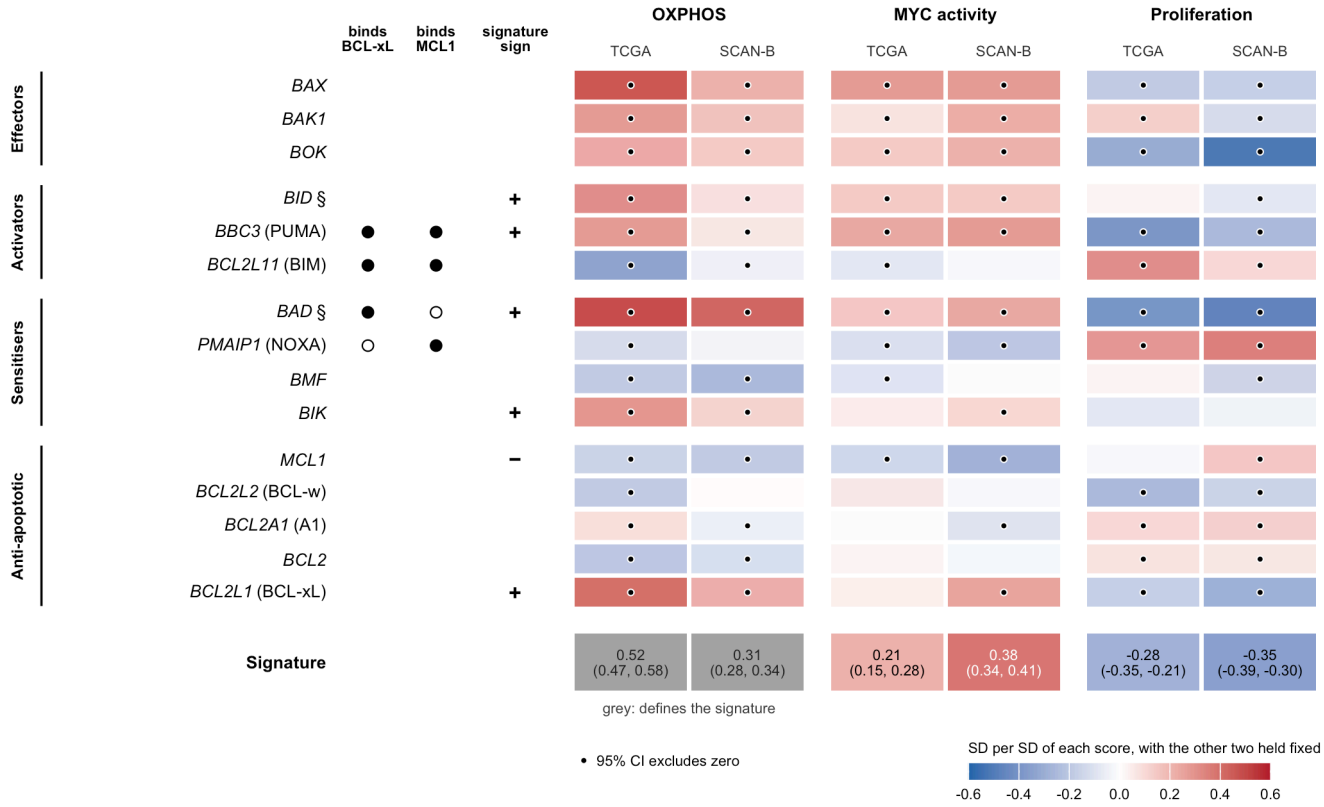
STYLE.md metrics

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Abstract	2	44	22.0	22	20–24	0 of 2	0	0
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Clinical, edited sentences	3	65	21.7	24	13–28	1 of 3	0	0
Closing, human sentence	1	33	33.0	33	33–33	0 of 1	0	0
Methods notes	3	78	26.0	24	23–31	0 of 3	0	0

The three Results paragraphs together: 18 sentences, 418 words, mean 23.2, median 23, range 14–34, 6 of 18 (33%) under 20 words. **That share is below STYLE.md's 35% floor**; no sentence was shortened for the metric, and whether to split one is the author's call.

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then version (b)'s addition (16 words). As the unmarked version, with a dot on each cell whose 95% confidence interval excludes zero.

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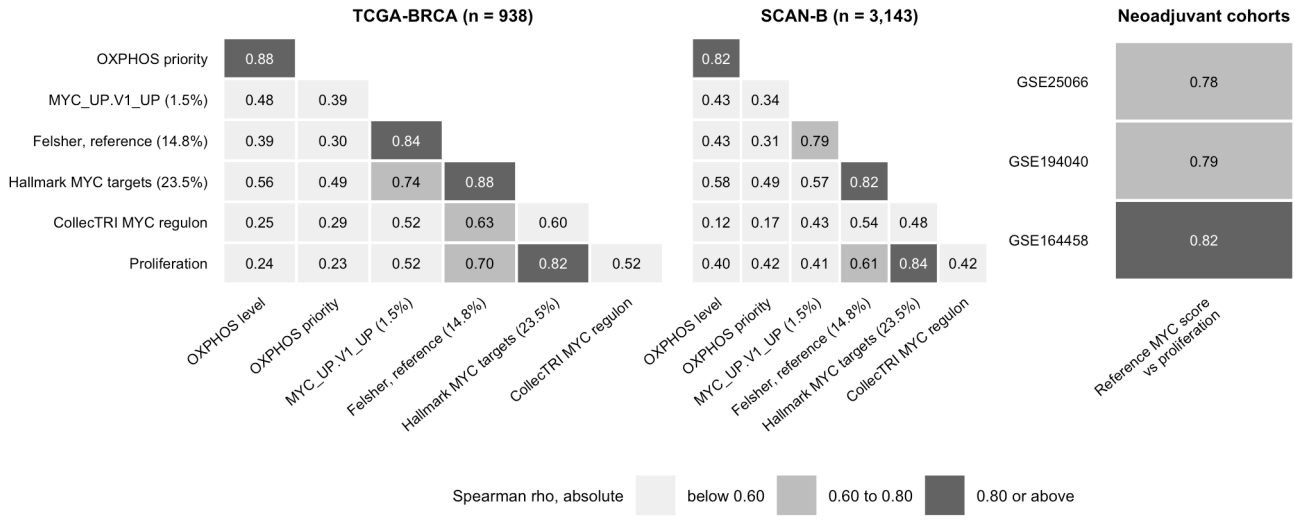
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The Extended Data panels

These are reproduced at final size, each with its legend draft verbatim from the script that drew it. The letters are placeholders: a to c are E48′s, e to k are E49′s. E48′s ED-d is superseded for the paper by ED-i, its extension; E48′s S-1 is kept as it was and is not repeated here.

ED-a (E48): the scores' correlations

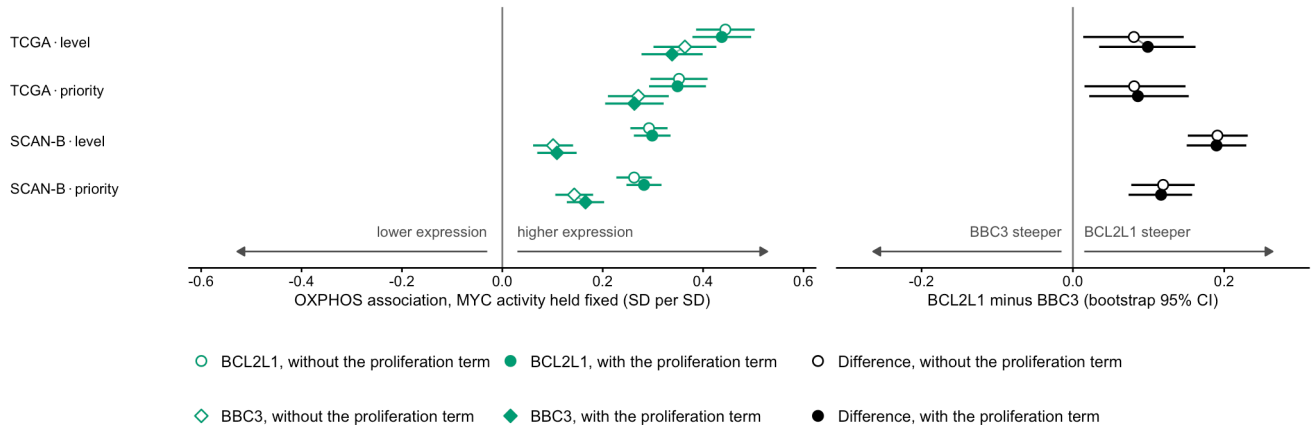
Cited as: Extended Data Fig. Xa, in sentences S2b and S6.



Legend draft (66 words). Spearman correlations among the two OXPHOS scores, four MYC activity scores and proliferation in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), shaded by band (below 0.60; 0.60 to 0.80; 0.80 or above). Each MYC activity score is labelled with the percentage of its genes in proliferation gene sets (Hallmark E2F targets and G2M checkpoint). Right, the reference MYC activity score against proliferation in three neoadjuvant cohorts.

ED-b (E48): *BCL2L1* against *BBC3*

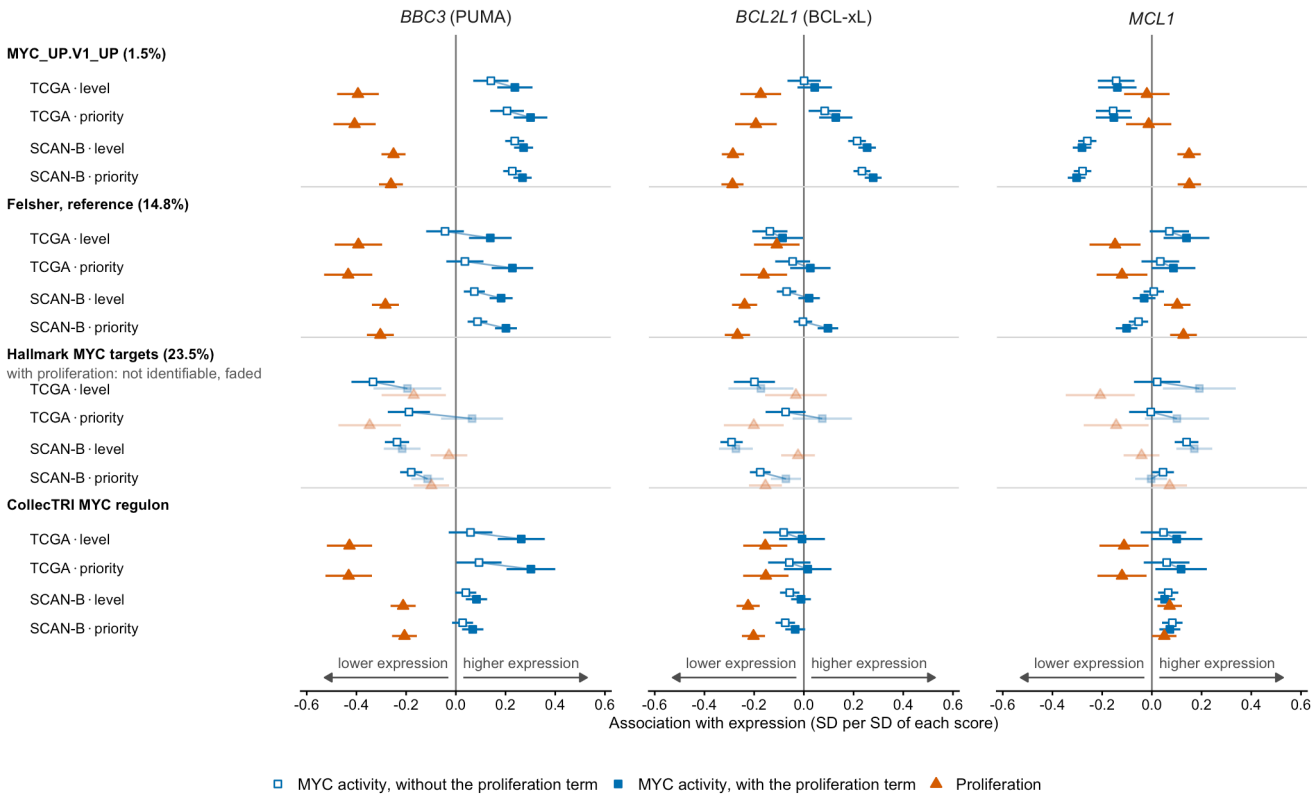
Cited as: Extended Data Fig. Xb, in sentence B1.



Legend draft (51 words). OXPHOS coefficients for BCL2L1 and BBC3 with MYC activity held fixed, without (open) and with (filled) the proliferation term, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), and the difference between them, with 95% percentile intervals from 2,000 paired bootstrap resamples of tumours. Bars on the coefficients, model-based 95% confidence intervals.

ED-c (E48): MYC activity on four scores, three genes

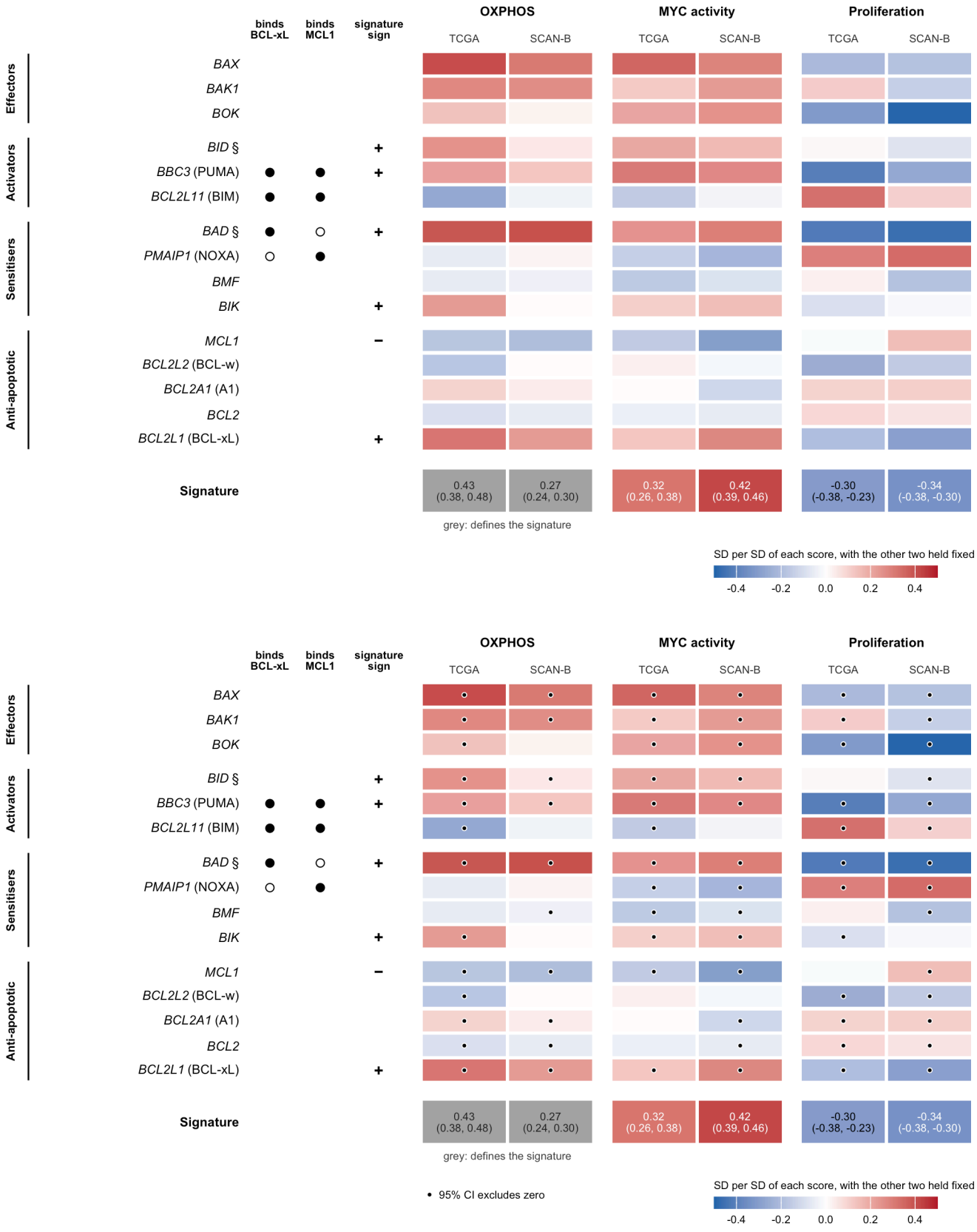
Cited as: not cited in the text. ED-f sits beside it (amendment item 3).



Legend draft (70 words). MYC activity coefficients for BBC3, BCL2L1 and MCL1 on four MYC activity scores, ordered by the percentage of their genes in proliferation gene sets, without (open) and with (filled) the proliferation term, with the proliferation coefficient (triangles), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Faded points are cells in which the score and proliferation correlate at 0.80 or more; they are shown and not interpreted. Bars, 95% confidence intervals.

ED-e: Fig. 4g on OXPHOS priority, versions (a) and (b)

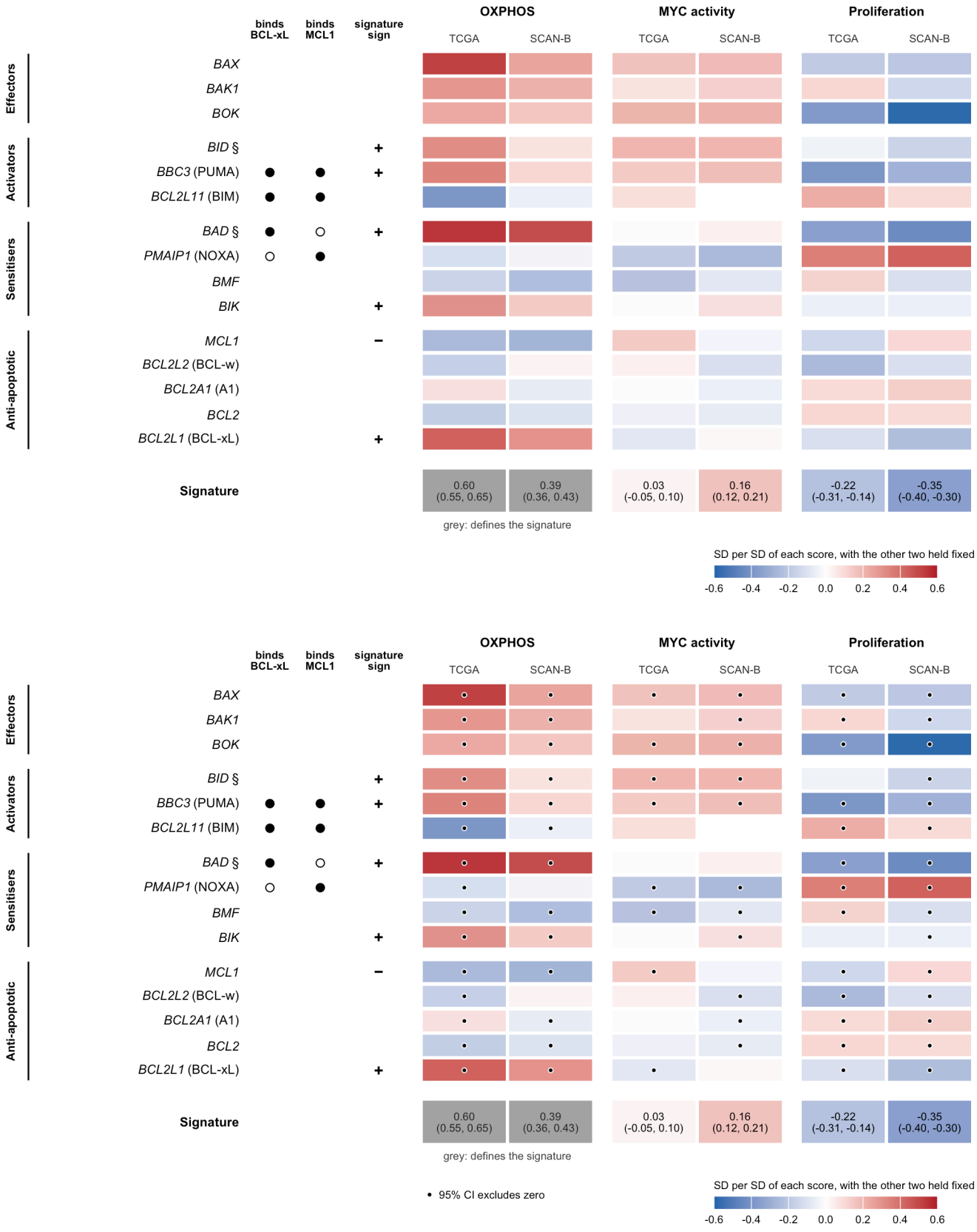
Cited as: not cited in the text; the 4g legend points to it.



Legend draft (14 words). As Fig. 4g, with OXPHOS scored as its priority within the mitochondrial programme (mitoPPS).

ED-f: Fig. 4g from the reference model, versions (a) and (b)

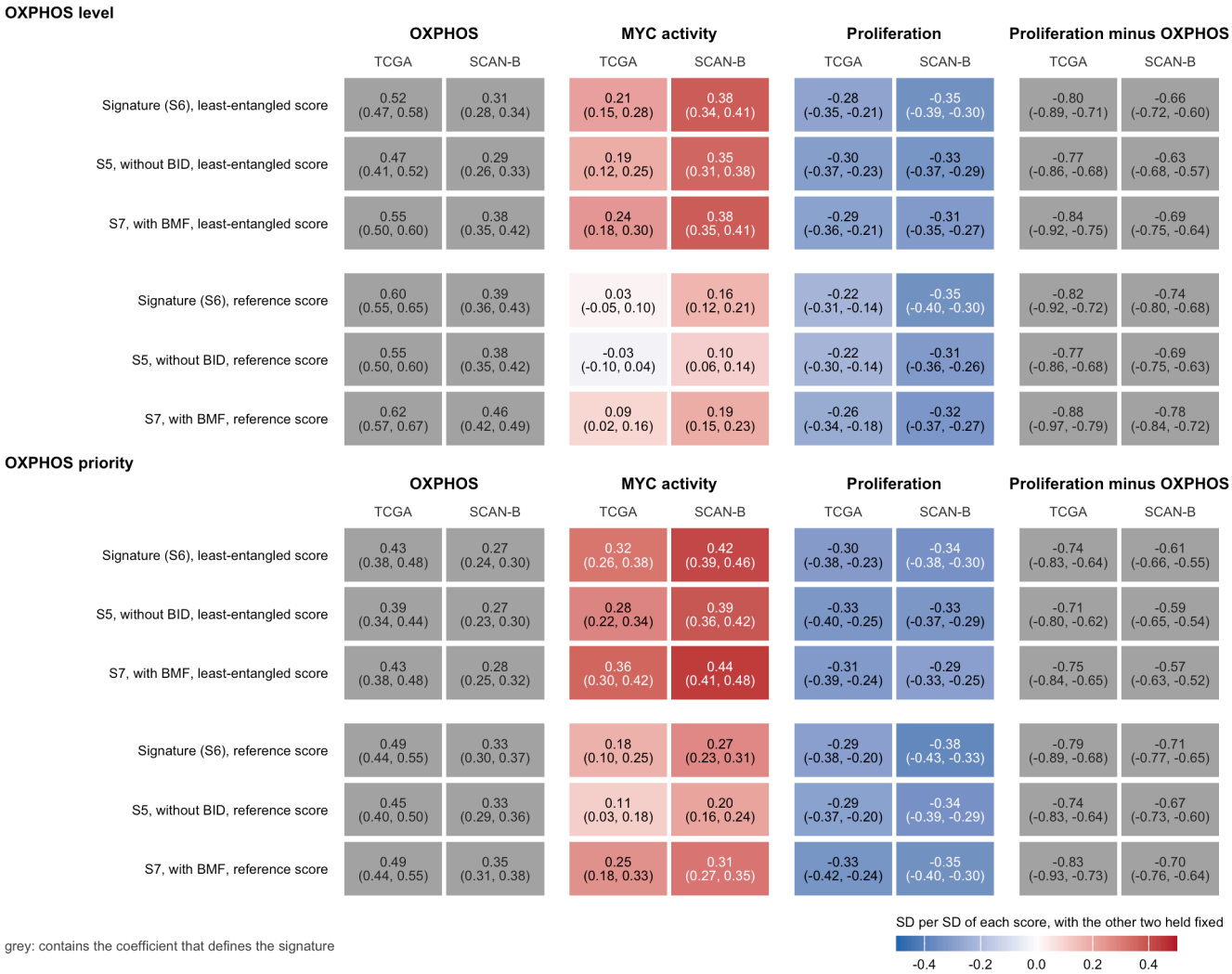
Cited as: not cited in the text; the 4g legend points to it.



Legend draft (20 words). As Fig. 4g, from models using the reference MYC activity score (Felsher), which correlates with proliferation at 0.61 to 0.70.

ED-g: the signature and its two variants

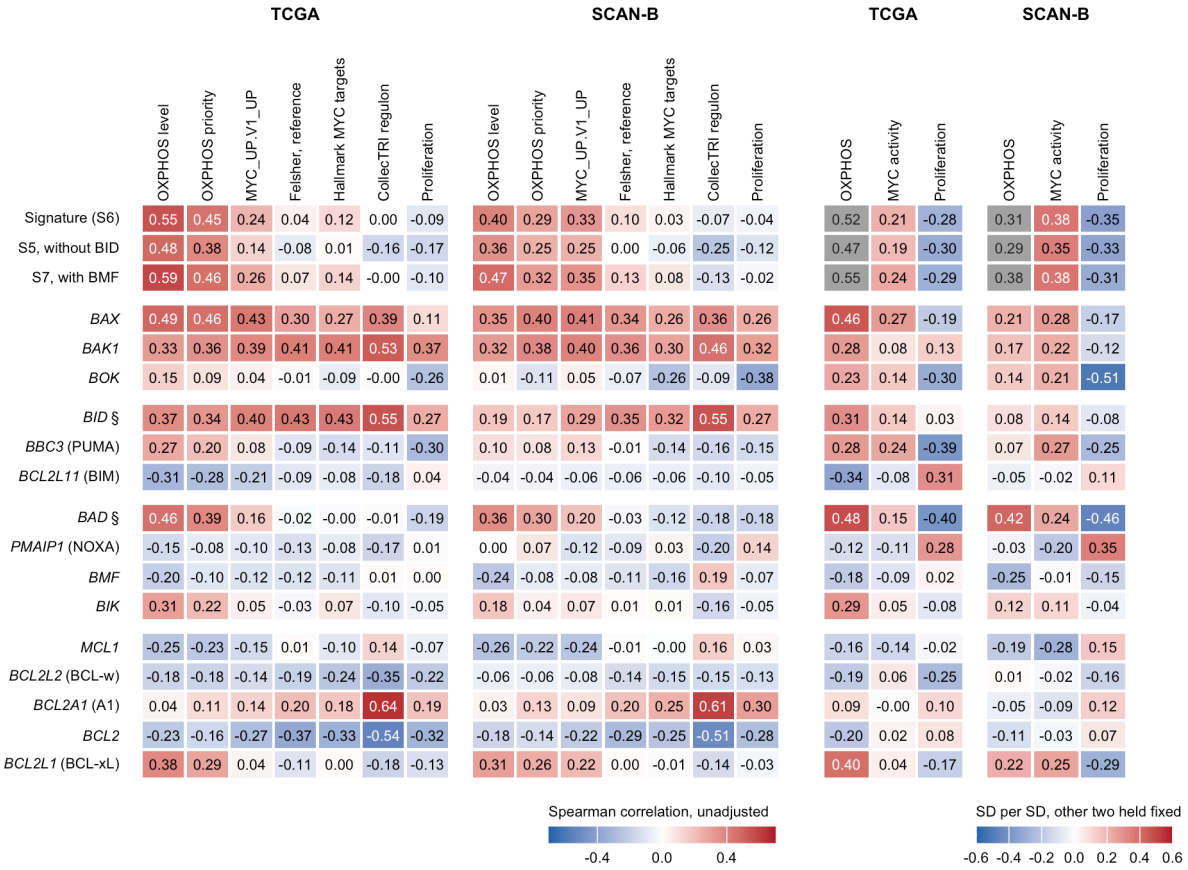
Cited as: not cited in the text.

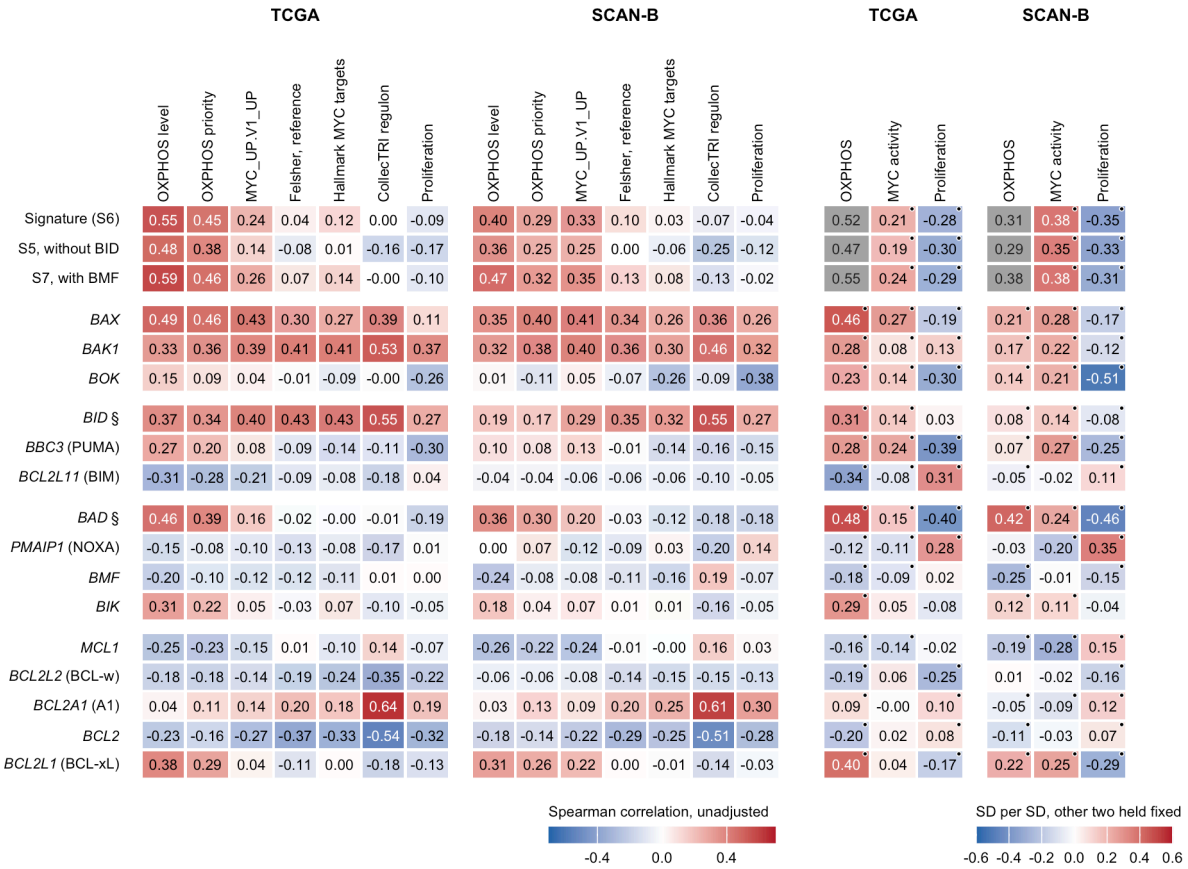


Legend draft (74 words). The six-gene signature (S6) and its two variants, S5 (without BID) and S7 (with BMF), on the MYC activity score with few proliferation genes and on the reference score, with OXPHOS scored as level and as priority, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each cell gives the coefficient (SD per SD) and its 95% confidence interval. Grey cells contain the OXPHOS coefficient, which defines the signature: OXPHOS itself, and proliferation minus OXPHOS.

ED-h: raw correlations beside the adjusted coefficients, versions (a) and (b)

Cited as: Extended Data Fig. Xh, in sentence S6.

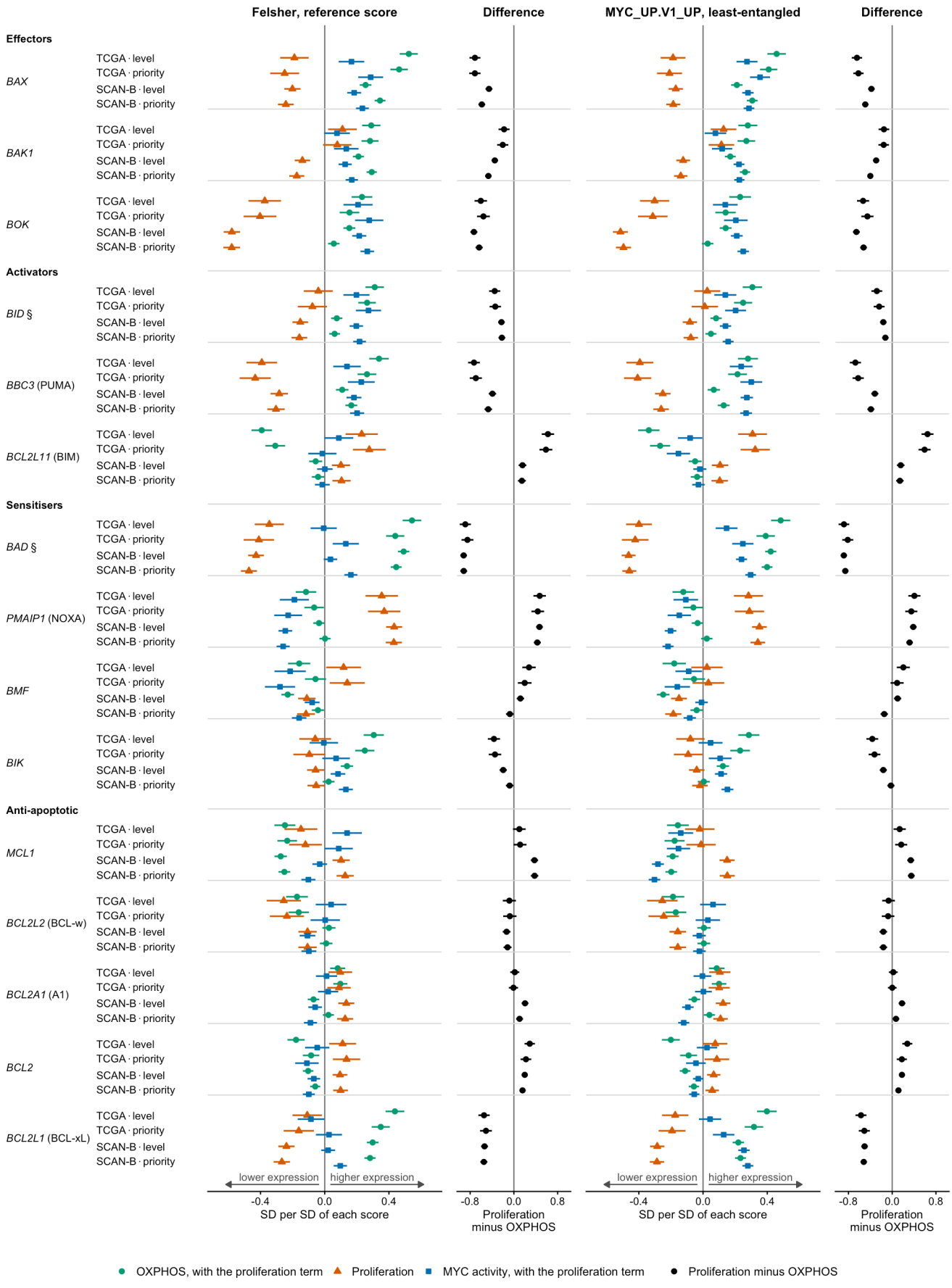




Legend draft (66 words). Left, unadjusted Spearman correlations of the signature, its variants and fifteen BCL-2 family genes with OXPHOS (level and priority), four MYC activity scores and proliferation, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Right, the adjusted coefficients of Fig. 4g. Grey, the signatures' association with OXPHOS, which defines them. The marked version adds a dot where the 95% confidence interval of an adjusted coefficient excludes zero.

ED-i: all fifteen genes, three coefficients and the difference

Cited as: not cited in the text.

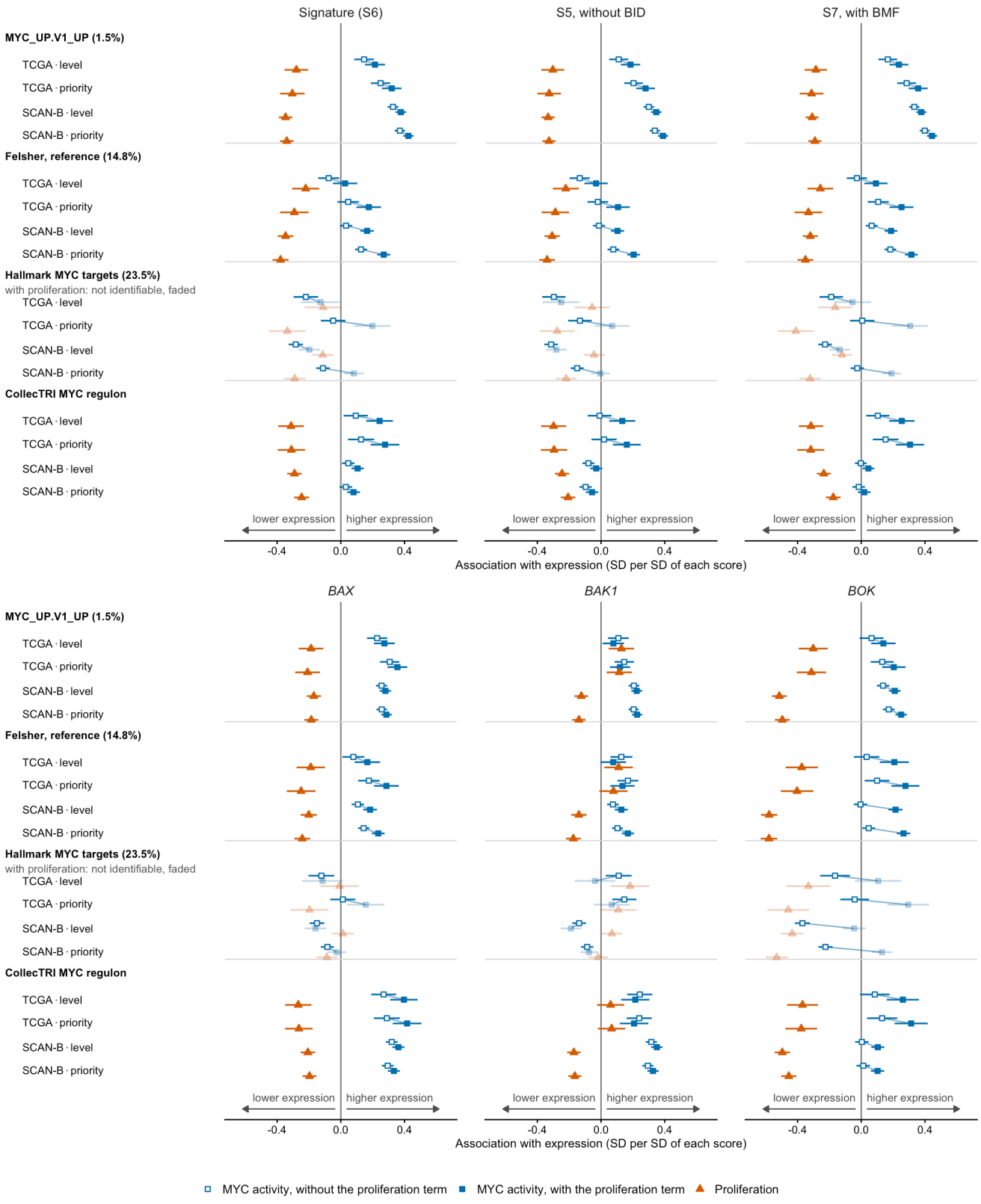


Legend draft (81 words). Fifteen BCL-2 family genes, in the groups of Fig. 1g: coefficients of OXPHOS (circles), proliferation (triangles) and MYC activity (squares), each with the other two held fixed, and proliferation minus OXPHOS, on the reference MYC activity score (left) and the score with few proliferation genes (right), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). § BH3-only proteins whose

activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). No correction for multiple genes. Bars, 95% confidence intervals.

ED-j: MYC activity on four scores, the signatures and the effectors

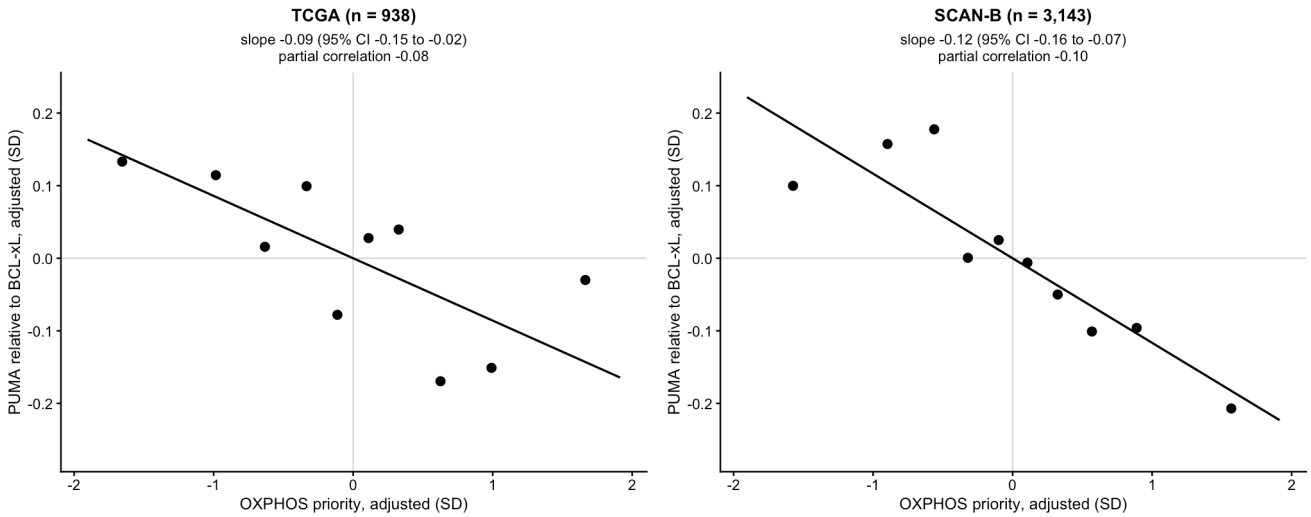
Cited as: Extended Data Fig. Xj, in sentence S7.



Legend draft (76 words). MYC activity coefficients for the signature, its two variants and BAX, BAK1 and BOK on four MYC activity scores, ordered by the percentage of their genes in proliferation gene sets, without (open) and with (filled) the proliferation term, with the proliferation coefficient (triangles), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Faded points are cells in which the score and proliferation correlate at 0.80 or more; they are shown and not interpreted. Bars, 95% confidence intervals.

ED-k: PUMA relative to BCL-xL against OXPHOS priority

Cited as: Extended Data Fig. Xk, in sentence B2.



Legend draft (61 words). BBC3 (PUMA) relative to BCL2L1 (BCL-xL) against OXPHOS priority, each adjusted for the reference MYC activity score, proliferation and intrinsic subtype (and purity and leukocyte fraction in TCGA-BRCA), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Points, means of the ten deciles of adjusted OXPHOS priority; line, the fitted slope, with its 95% interval from 2,000 paired bootstrap resamples of tumours.